

FerroEnrich User Manual

1. Overview

FerroEnrich is a web-based R Shiny platform for integrated transcriptomic analysis of ferroptosis and senescence programs. The tool is designed to help researchers analyze RNA-seq count data, identify differentially expressed genes, calculate ferroptosis-associated activity, explore liver ferroptosis and senescence modules, visualize interaction networks, and evaluate ferroptosis–senescence cross-talk.

FerroEnrich supports both **human** and **mouse** datasets and is optimized for liver disease–related transcriptomic studies such as NASH, fibrosis, cholestasis, cirrhosis, hepatocellular carcinoma, and metabolic liver disease.

2. Input Data Requirements

2.1 Count Matrix

The count matrix should contain **raw RNA-seq read counts**.

Expected format:

Gene ID	Sample1	Sample2	Sample3
ENSG00000141510	120	95	210
ENSG00000171862	50	60	48

Requirements:

- First column should contain gene IDs.
- Gene IDs can be Ensembl IDs such as ENSG... or ENSMUSG....
- Remaining columns should be sample names.
- Counts should be raw integer-like read counts.
- Missing values are not recommended. FerroEnrich attempts to replace missing values with zero during DESeq2 preparation.

Accepted file formats:

.csv

.tsv

.txt

.xlsx

.xls

2.2 Metadata File

The metadata file should contain sample annotation information.

Expected format:

SampleID	Type
Sample1	Control
Sample2	NASH
Sample3	NASH

Requirements:

- One row per sample.
- A column containing sample IDs.
- A group/condition column for comparison.
- Sample names in metadata must match column names in the count matrix.

Common metadata columns:

SampleID

Description

Type

Group

Condition

Treatment

Disease

3. Sidebar Input Panel

The sidebar contains the main data input controls.

3.1 Data Type

Currently, FerroEnrich uses:

Read Counts

The app expects raw RNA-seq counts for DESeq2-based differential expression analysis.

3.2 Load Demo Data

Two built-in demo datasets are available:

Load Demo Data 1

Loads:

NASH_1

Metadata_1

This can be used to test the app quickly.

Load Demo Data 2

Loads:

NASH_2

Metadata_2

This provides an additional example dataset.

3.3 Clear Data

The **Clear Data** button removes currently loaded count and metadata files from the app. Use this before uploading a new dataset.

3.4 Upload Counts

Use this option to upload the RNA-seq count matrix.

The first column should contain gene IDs, and all other columns should be sample counts.

3.5 Upload Metadata

Use this option to upload the sample metadata file.

After uploading metadata, select:

Sample ID column

Group column

Reference level

3.6 Sample ID Column

This tells FerroEnrich which metadata column contains sample IDs.

Example:

Description
SampleID
ID

The selected sample IDs must match count matrix column names.

3.7 Group Column

This defines the biological or experimental groups used in DESeq2 analysis.

Example:

Control vs NASH
Chow vs NASH Diet
WT vs KO
Healthy vs Disease

3.8 Reference Level

This is the baseline/control group used for differential expression.

Example:

Control
Chow
WT
Healthy

DESeq2 results are interpreted relative to this reference group.

3.9 Species

FerroEnrich supports:

Human
Mouse

The app attempts to auto-detect species from Ensembl IDs:

ENSG... = Human
ENSMUSG... = Mouse

You can manually override this selection.

4. Load Data Tab

The **Load Data** tab provides an overview of the uploaded dataset.

4.1 About FerroEnrich

This section describes the purpose of the tool and its main analysis capabilities.

4.2 Counts Preview

Displays the first rows of the uploaded count matrix.

Use this to confirm:

- Gene IDs are present.
- Sample names are correct.
- Count values are loaded properly.

4.3 Metadata Preview

Displays the first rows of the metadata file.

Use this to confirm:

- Sample IDs are present.
- Group labels are correct.
- Metadata columns are properly recognized.

4.4 Summary

The summary section reports:

Number of genes

Number of samples

Metadata rows

Selected group column

Group levels

Sample IDs

Gene IDs

This is useful for checking whether counts and metadata are correctly aligned.

5. DEG Tab

The **DEG** tab performs differential gene expression analysis using DESeq2.

5.1 Run DESeq2

Click:

Run DESeq2

The app will:

1. Align count matrix and metadata.
2. Relevel the group factor based on selected reference.
3. Convert counts to integer mode.
4. Replace missing count values with zero.
5. Remove genes with all zero counts.
6. Filter low-count genes.
7. Run DESeq2.
8. Map Ensembl IDs to gene symbols.
9. Generate a final DEG table.

5.2 DEG Design

The DESeq2 model uses:

design = ~ group

The comparison is performed as:

test group vs reference group

Example:

NASH vs Control

Disease vs Healthy

KO vs WT

5.3 Volcano Settings

The volcano plot can be customized using:

Significance metric

Options:

padj

pvalue

Recommended:

padj

Significance cutoff

Default:

0.05

Genes with values below this cutoff are considered significant.

log2FC cutoff

Default:

1

Genes with absolute log2 fold change above this cutoff are highlighted as upregulated or downregulated.

Label top N genes

Controls the number of top significant genes labeled on the volcano plot.

Example:

20

Point size

Controls the size of volcano plot points.

Label size

Controls the size of gene labels.

Axis title size and axis text size

Controls volcano plot font sizes.

5.4 Custom Gene List

The custom gene list now accepts **gene symbols only**.

Example:

GPX4, SLC7A11, ACSL4, FTH1, TFRC

or one gene per line:

GPX4
SLC7A11
ACSL4
FTH1
TFRC

The app treats all custom genes as:

Symbols

No Ensembl ID upload or list type selection is required.

5.5 Highlight Prone/Resistant

This option controls whether ferroptosis-prone and ferroptosis-resistant genes are highlighted on the volcano plot.

When enabled:

- Ferroptosis-prone genes are highlighted.
- Ferroptosis-resistant genes are highlighted.
- Custom pasted symbols are also highlighted.

5.6 Volcano Plot

The volcano plot displays:

x-axis = $\log_2$ Fold Change

y-axis = $-\log_{10}(\text{padj or pvalue})$

Color meaning:

Red = Upregulated genes

Blue = Downregulated genes

Grey = Not significant genes

Additional markers:

Black circle = custom pasted gene symbol

Red triangle = matched ferroptosis-prone gene

Blue triangle = matched ferroptosis-resistant gene

5.7 Download Volcano Plot

Click:

Download Volcano (PNG)

This saves the volcano plot as a high-resolution PNG file.

5.8 Matched Ferroptosis Genes

FerroEnrich identifies matched ferroptosis genes using backend gene lists:

prone_genes

resistant_genes

The matching is based on **gene symbols**.

The app reports:

Number of matched prone genes

Number of matched resistant genes

5.9 Matched Prone Genes Table

This table shows significantly upregulated genes that match the ferroptosis-prone gene list.

Columns include:

Gene

log2FoldChange

pvalue

padj

FerroScore

Download option:

Download matched Prone

5.10 Matched Resistant Genes Table

This table shows significantly downregulated genes that match the ferroptosis-resistant gene list.

Columns include:

Gene

log2FoldChange

pvalue

padj
FerroScore

Download option:

Download matched Resistant

6. Ferroptosis Index Value

The DEG tab includes the **Ferroptosis Index Value**, or **FIV**.

6.1 Purpose of FIV

FIV summarizes dataset-level ferroptosis tendency by integrating:

log2FoldChange
FerroScore
Prone/resistant direction

The score reflects whether the transcriptomic profile is shifted toward ferroptosis-prone or ferroptosis-resistant activity.

6.2 FIV Calculation

For matched genes, FerroEnrich calculates:

$\text{WeightedExpression} = \log2\text{FoldChange} \times \text{FerroScore}$

Then:

$\text{Sum_Prone} = \text{sum weighted expression of prone genes}$
 $\text{Sum_Resistant} = \text{sum weighted expression of resistant genes}$
 $\text{Raw_FIV} = \text{Sum_Prone} - \text{Sum_Resistant}$

The raw FIV is scaled to a 0–5 range for visualization.

6.3 FIV Categories

FIV is classified as:

FIV Range Category

≤ 2.50 Normal

2.50–2.99 Mild

3.00–3.49 Moderate

FIV Range Category

3.50–4.49 High

≥ 4.50 Severe

6.4 FIV Gauge Plot

The gauge plot displays the scaled FIV value and category.

Download options:

Classic FIV (PNG)

Classic FIV (CSV)

The CSV contains:

Sum_Prone_Weighted

Sum_Resistant_Weighted

Raw_FIV

Scaled_FIV

Category

7. Matched Gene Heatmap

The DEG tab includes a heatmap of matched prone and resistant genes.

7.1 Input Genes

The heatmap uses:

Matched Prone genes U Matched Resistant genes

7.2 Expression Transformation

The app uses normalized expression values and row-wise Z-score scaling.

The heatmap helps visualize whether matched ferroptosis genes separate experimental groups.

7.3 Heatmap Options

Options include:

Cluster rows

Cluster columns

PNG width
PNG height

7.4 Download Heatmap

Click:

Download Heatmap (PNG)

This saves the matched gene heatmap.

8. DEG Table

The DEG table provides the full differential expression output.

Columns include:

ID
Gene
baseMean
log2FoldChange
lfcSE
stat
pvalue
padj

The table is filtered and ranked based on selected volcano settings.

Download option:

Download Results (CSV)

9. Pathways Tab

The **Pathways** tab evaluates curated ferroptosis and senescence modules using DEG results.

9.1 Dataset Selection

Available pathway/module datasets:

Liver_ferroptosis
Liver_senescence

9.2 Module Selection

You can select:

All

Specific module

Example ferroptosis modules:

Hepatic_Ferroptosis_Core

PUFA_Peroxidation_and_Phospholipid_Assembly

Iron_Handling_and_Ferritinophagy

CoQ10_FSP1_Axis

NRF2_KEAP1_Antioxidant_Response

p53_SAT1_Lipoxygenase_Link

Example senescence modules:

Cell_Cycle_Arrest_p53_p21_p16_pRB

DNA_Damage_Response_DDR_Telomere

SASP_Cytokines_Chemokines

SASP_Proteases_ECM_Remodeling

Hepatic_Stellate_Cell_Senescence_and_Fibrosis

9.3 Pathway Significance Settings

Controls include:

Significance metric: padj or pvalue

padj/pvalue cutoff

|log2FC| cutoff

9.4 Run Pathway Analysis

Click:

Run Pathway Analysis

The app tests whether selected modules contain enriched upregulated or downregulated significant genes.

9.5 Enriched Modules Table

The output table reports:

Direction

Adjusted P value

Number of genes

Module

Direction includes:

Up regulated

Down regulated

Download option:

Download Summary (CSV)

9.6 Matched Genes in Selected Module

This table shows genes from the selected module that are present in the DEG table.

Columns include:

Gene

log2FoldChange

pvalue

padj

Module

Download option:

Download Genes (CSV)

9.7 Module Heatmap

The module heatmap visualizes expression of genes in the selected pathway module.

Options include:

Cluster rows

Cluster columns

PNG width

PNG height

Download option:

Download Module Heatmap (PNG)

10. Enrichment Tab

The **Enrichment** tab performs GSEA-style pathway enrichment.

10.1 Dataset Selection

Available datasets:

Liver_ferroptosis
Liver_senescence

10.2 Module Selection

You can run enrichment for:

All modules
Specific module

10.3 Ranking Metric

Options:

padj
pvalue

The app builds a ranked gene list based on direction and significance.

10.4 GMT File Option

The option:

Use GMT file if available

allows FerroEnrich to use external GMT gene set files if present.

If GMT files are unavailable, the app uses internal CSV or hard-coded module definitions.

10.5 Run GSEA

Click:

Run GSEA

The app uses clusterProfiler::GSEA to calculate enrichment.

10.6 GSEA Results Table

The output includes:

Module
Description
setSize
enrichmentScore
NES
pvalue
padj

qvalue
rank
leading_edge
core_enrichment
Dataset

Important columns:

NES > 0 = enriched in upregulated genes
NES < 0 = enriched in downregulated genes
padj = adjusted enrichment significance
core_enrichment = leading-edge genes

Download option:

Download GSEA results (CSV)

10.7 Enrichment Plots

Plot types:

Dot
Bar
Lollipop

Color options:

$-\log_{10}(\text{padj})$
 $-\log_{10}(\text{pvalue})$

The app displays separate plots for:

Upregulated modules
Downregulated modules

Download options:

Download Upregulated (PNG)
Download Downregulated (PNG)

11. Network Tab

The **Network** tab visualizes gene interaction networks for selected modules.

11.1 Gene Module Dataset

Available options:

Liver ferroptosis
Senescence

11.2 Module Selection

You can select:

All
Specific module

The network is generated from genes in the selected module.

11.3 Network Source

Available sources:

STRING PPI
Complete graph

STRING PPI

Uses STRING protein-protein interactions.

Recommended for biological interaction networks.

Complete graph

Connects all selected genes together.

Useful as a fallback when STRING mapping is incomplete or unavailable.

11.4 STRING Score Cutoff

For STRING PPI networks, the score cutoff controls interaction confidence.

Example:

400

Higher values produce more stringent networks.

11.5 Max Edges

Controls the maximum number of network edges displayed.

Example:

300

Use smaller values for faster visualization.

11.6 Run Network

Click:

Run Network

The app builds the selected network.

11.7 Interactive Network Viewer

The network is displayed using visNetwork.

Features include:

Zoom

Drag nodes

Move network

Select nodes

Highlight nearest neighbors

Relayout

Save network PNG from browser

11.8 Node Color

Nodes are colored by log2 fold change.

General interpretation:

Blue = lower/downregulated expression

White = near zero

Red = higher/upregulated expression

11.9 Network Downloads

Available downloads:

Download network table

Download current network PNG

Download network table

Exports network edges as CSV.

Download current network PNG

Downloads the currently selected network source:

STRING PPI

or

Complete graph

Important workflow:

1. Select network source.
2. Click Run Network.
3. Click Download current network PNG.

For exact zoomed browser view, use the built-in:

Save network PNG

inside the interactive network.

12. Cross-talk Tab

The **Cross-talk** tab evaluates interactions between ferroptosis modules and senescence modules.

12.1 Purpose

This tab helps identify whether ferroptosis-related transcriptional programs are positively or negatively associated with senescence/fibrosis modules.

It integrates:

Literature-based score matrix
Module expression correlation
Combined cross-talk score

12.2 Matrix to Display

Options:

Literature
Correlation
Combined

Literature

Displays predefined biological interaction scores between ferroptosis and senescence modules.

Score range:

-2 to +2

Positive scores indicate pro-ferroptotic cross-talk.

Negative scores indicate protective or anti-ferroptotic cross-talk.

Correlation

Displays correlation between module activity scores across samples.

Combined

Calculated as:

Combined = Literature score × Correlation

This is the recommended view for biological interpretation.

12.3 Module Ordering

Rows and columns can be ordered by:

Input order

Hierarchical clustering

Hierarchical clustering groups modules with similar cross-talk patterns.

12.4 Run Cross-talk Analysis

Click:

Run Cross-talk Analysis

The app will:

1. Load ferroptosis and senescence modules.
2. Normalize expression data.
3. Map Ensembl IDs to gene symbols.
4. Calculate module activity scores.
5. Compute module-module correlations.
6. Build literature, correlation, and combined matrices.
7. Generate FSI plots.
8. Generate top cross-talk edges.

9. Build module network.

13. Cross-talk Heatmap

The heatmap visualizes module-level cross-talk.

13.1 Heatmap Interpretation

Red = pro-ferroptotic cross-talk

Blue = protective / anti-ferroptotic cross-talk

White = weak or neutral cross-talk

Rows:

Ferroptosis modules

Columns:

Senescence modules

13.2 Heatmap Downloads

Available downloads:

Download heatmap (PNG)

Download heatmap (PDF)

Download matrix (CSV)

The CSV file contains the numerical matrix values.

14. Ferroptosis Susceptibility Index

The **Ferroptosis Susceptibility Index**, or **FSI**, estimates sample-level ferroptosis susceptibility based on module activity.

14.1 FSI Formula

FSI is calculated as:

$$\text{FSI} = \text{mean}(\text{z-score of pro-ferroptotic modules}) \\ - \text{mean}(\text{z-score of protective modules})$$

Higher FSI suggests greater ferroptosis susceptibility.

Lower FSI suggests stronger protective ferroptosis resistance.

14.2 FSI Grouping

The user can select a metadata column to group samples.

Examples:

Type

Group

Condition

Treatment

Disease

Status

14.3 FSI Distribution Plot

This plot shows the overall distribution of FSI values across all samples.

14.4 FSI by Group Plot

This plot compares FSI values across metadata groups.

When enabled, the plot shows:

Violin plot

Boxplot

Individual sample points

14.5 FSI Downloads

Available downloads:

Download FSI values (CSV)

Download current FSI PDF

Download FSI values

Exports per-sample FSI values.

Download current FSI PDF

Saves the current FSI distribution and FSI by group plots together in one PDF.

15. Top Ferroptosis–Senescence Cross-talk Pairs

This table ranks module pairs by combined cross-talk strength.

15.1 Show Top N Edges

Controls how many module pairs are displayed.

Example:

15.2 Direction Filter

Options:

All

Pro-ferroptotic only

Protective only

Pro-ferroptotic

Positive combined score.

Protective

Negative combined score.

15.3 Top Edge Table Columns

The table includes:

Ferro

Sen

Literature

Correlation

Combined

absCombined

Direction

Download option:

Download shown edges (CSV)

16. Cross-talk Module Network

The module network visualizes top ferroptosis–senescence module pairs.

16.1 Network Nodes

Nodes represent modules.

Node types:

Ferroptosis modules

Senescence modules

16.2 Network Edges

Edges represent cross-talk relationships.

Edge color:

Red = pro-ferroptotic edge

Blue = protective edge

Edge width reflects the strength of the combined cross-talk score.

16.3 Network Layout

Available layouts:

Fruchterman-Reingold

Circle

Kamada-Kawai

If negative weights are present, FerroEnrich uses positive layout weights internally while preserving signed values for edge color.

16.4 Download Network

Download option:

Download network (PDF)

This exports the cross-talk module network.

17. Manual Tab

The **Manual** tab displays the FerroEnrich user manual directly inside the app.

The manual should be saved in the working directory as:

Manual.pdf

The app accesses it using:

assets/Manual.pdf

If the PDF does not load inside the browser, use:

Open Manual PDF

to open it in a new tab.

18. Recommended Workflow

A typical FerroEnrich analysis should follow this order:

Step 1: Load data

Upload:

Counts file

Metadata file

or use demo data.

Step 2: Select metadata settings

Choose:

Sample ID column

Group column

Reference level

Species

Step 3: Run DESeq2

Go to DEG tab and click:

Run DESeq2

Step 4: Inspect DEG outputs

Check:

Volcano plot

DEG table

Matched prone genes

Matched resistant genes

FIV score

Matched gene heatmap

Step 5: Run pathway analysis

Go to Pathways tab and click:

Run Pathway Analysis

Step 6: Run enrichment analysis

Go to Enrichment tab and click:

Run GSEA

Step 7: Build network

Go to Network tab:

Select module

Select STRING PPI or Complete graph

Click Run Network

Download current network PNG

Step 8: Run cross-talk analysis

Go to Cross-talk tab and click:

Run Cross-talk Analysis

Then examine:

Cross-talk heatmap

FSI distribution

FSI by group

Top module-pair table

Crosstalk module network

19. Troubleshooting

19.1 No overlapping sample IDs

Cause:

Sample names in counts and metadata do not match.

Fix:

Make sure count matrix column names match the selected metadata sample ID column.

19.2 Too few genes remain after filtering

Cause:

Count matrix may not contain raw RNA-seq counts or may have too many low-count genes.

Fix:

Use raw counts and check that samples contain sufficient read depth.

19.3 No matched ferroptosis genes

Possible causes:

- Backend gene lists are missing.

- Gene symbols did not map correctly.
- No significant genes match prone/resistant lists.
- Wrong species selected.

Fix:

Check:

prone_genes.csv

resistant_genes.csv

species setting

gene symbol mapping rate

19.4 Volcano custom genes not highlighted

Cause:

Custom list accepts **Symbols only**.

Fix:

Use symbols such as:

GPX4

SLC7A11

ACSL4

Do not use Ensembl IDs in the custom gene box.

19.5 Network is slow

Possible causes:

- Too many genes.
- Complete graph creates many edges.
- STRING query is large.

Fix:

Reduce:

Max edges

or select a smaller module.

19.6 STRING network has few edges

Possible causes:

- Genes failed to map to STRING.
- STRING score cutoff is too high.
- Selected module has few known interactions.

Fix:

Lower STRING score cutoff or use Complete graph.

19.7 Cross-talk network layout error

Cause:

Some combined scores are negative.

Fix:

Use updated code that uses absolute weights for layout and signed weights for edge color.

19.8 App shows “argument is of length zero”

Common cause:

Server code refers to removed UI inputs such as:

```
input$gene_list_type  
input$ferro_list_type
```

Fix:

Since the app now uses symbols only, replace those inputs with:

"Symbols"

or remove conditional Ensembl/Symbol logic.

19.9 App sourcing error: missing comma

Cause:

A UI element is missing a comma between components.

Example wrong:

```
)  
div(...)
```

Correct:

```
),  
div(...)
```

20. Output Files Generated by FerroEnrich

FerroEnrich can generate:

- DESeq2 results CSV
- Volcano plot PNG
- Matched prone genes CSV
- Matched resistant genes CSV
- FIV summary CSV
- FIV gauge PNG
- Matched gene heatmap PNG
- Pathway summary CSV
- Pathway matched genes CSV
- Module heatmap PNG
- GSEA results CSV
- GSEA upregulated plot PNG
- GSEA downregulated plot PNG
- Network edge table CSV
- Current network PNG
- Cross-talk heatmap PNG
- Cross-talk heatmap PDF
- Cross-talk matrix CSV
- FSI values CSV
- Current FSI PDF
- Top cross-talk edges CSV
- Cross-talk module network PDF